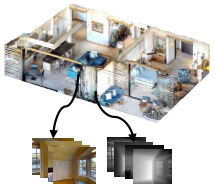
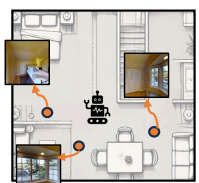
Instruction: Walk past the kitchen and go behind the couch and take a right into the fitness room. Stop next to the treadmill.



Waypoint Prediction



Candidate Direction 1:

Scene Objects:

chair | computer | computer desk | computer monitor | table …

Spatial Scene Description:

I can see a room contains a desk, a chair, a computer tower, a bucket, and two suitcases.
The objects are positioned as follows:
- The desk is in the foreground, closest to the camera.
- The chair is to the left of the desk.
- The computer tower is on the desk, slightly behind the chair…

The distances from the camera are as follows:
- The desk is approximately 0.25 meter away.
- The chair is approximately 0.5 meters away.
- The computer tower is approximately 0.65 meters away…

Candidate Direction 2:

…

Instruction Comprehension:

Decomposed Actions: 1. Walk past the kitchen; 2. Go behind the couch; 3. Take a right into the fitness room; 4. Stop next to the treadmill
Extracted Landmarks: kitchen, couch, fitness room, treadmill

Progress Estimation:

Summarised Trajectory: Step 1 Observation: Direction Left, range(left 90 to left 120) The room is large and empty with …
Executed Actions Thought: Based on the navigation history and current observation, the following actions have likely been executed:
None

Decision Making:

Thoughts: The first action in the instruction is to walk through the kitchen. However, none of the current environment descriptions mention the kitchen. Therefore, I need to find the kitchen first. Looking at the scene description, Direction 1 is a study room with computer, Direction 2 depicts a living room with a dining area, while direction 3 shows a living room with a TV. Since the kitchen is usually located near the dining area, I would consider moving in Direction 2. Therefore, I decide to move in Direction 2.
Prediction: Direction 2

Candidate Waypoint

Scene Perception

RAM | SpatialBot

Textual Observation

LLM Navigator

Open-Source LLM

Instruction

Textual History Trajectory

Action

